

Session 1 — Vectors from Scratch and Motion in a Straight Line: The Kinematic Equations

Duration: 60 minutes. It splits cleanly into two 30-minute halves: vectors, then kinematics.

Position in the course: the first Physics session. Everything after it builds on this one: Session 2's projectiles, Session 4's relative velocity and Sessions 5–6's forces.

Grade-9 foundation: Class 9 Ch4 *Describing Motion Around Us*: distance vs displacement, speed vs velocity, acceleration, the equations of uniformly accelerated motion. *The Class 9 PDF is on the Mac and wasn't reachable today, so section numbers from it are left out.*

Grade-11 depth: Glencoe *Physics: Principles and Problems*:

- Ch2 “Representing Motion”: position, displacement, velocity
- Ch3 “Accelerated Motion”: §3.1 Acceleration, §3.2 Motion with Constant Acceleration, §3.3 Free Fall
- Ch5 §5.1 “Vectors”: adding vectors, and components

Hewitt *Conceptual Physics* adds Ch2 §2.1–2.6 (motion is relative, speed, velocity, acceleration, free fall “how fast” and “how far”) and Ch6 §6.1–6.4 (vector and scalar quantities, adding vectors geometrically).

$g = 9.81 \text{ m/s}^2$ throughout, matching the school.

Lesson Plan

Objectives. By the end of this session the student should be able to:

1. Tell **scalars** from **vectors**, add vectors tip-to-tail, and find a resultant's size and direction with Pythagoras and tangent.
2. **Resolve** a vector into x- and y-components, and add vectors by components.
3. Tell **distance** from **displacement** and **speed** from **velocity**, and find average speed, average velocity and acceleration.
4. **Derive** the kinematic equations for constant acceleration and choose the right one for a problem.
5. Solve **free-fall** problems with $a = -g$.

Time	Segment
0–5 min	Warm-up: “How far and which way?”
5–25 min	Part 1: vectors from scratch (scalars vs vectors, adding, components)
25–35 min	Part 2: describing motion (distance, displacement, speed, velocity, acceleration)
35–52 min	Part 3: the kinematic equations, derived one from another, with worked examples
52–57 min	Part 4: free fall
57–60 min	Exit ticket, assign homework

Pacing note. If time is short, keep the derivations in Part 3 (they are the heart of the session) and set component addition (Worked Example 3) and free fall as reading plus homework.

Warm-up (5 min)

1. You walk 3 km east, then 4 km north. How far did you walk? How far are you from where you started?
→ You **walked 7 km** (distance), but you're only **5 km** from the start (displacement). The difference

between “how far along the path” and “how far and which way” is the whole idea of vectors.

2. A car’s speedometer reads a steady 60 km/h as it goes around a bend. Is its velocity constant? → **No**.
The speed is constant but the direction changes, so the velocity changes. Velocity is a vector.

Formula Sheet — Session 1

Label	Equation	In words
V1	$R = \sqrt{A^2 + B^2}$, $\tan\theta = B/A$	resultant of two perpendicular vectors
V2	$A_x = A \cos\theta$, $A_y = A \sin\theta$	components (θ measured from the x-axis)
V3	$R_x = A_x + B_x$, $R_y = A_y + B_y$	add vectors by adding their components
K1	average speed = distance / time	a scalar
K2	$v_{avg} = \Delta x / \Delta t$ (displacement / time)	a vector
K3	$a = \Delta v / \Delta t = (v - u) / t$	acceleration: change in velocity per second
S0	$s = v_{avg} \times t = \frac{1}{2}(u + v)t$	distance/displacement = average velocity $\times$ time (constant a)
S1	$v = u + at$	final velocity
S2	$s = ut + \frac{1}{2}at^2$	displacement after time t
S3	$v^2 = u^2 + 2as$	links speed and displacement, with no time
FF	$a = -g = -9.81 \text{ m/s}^2$ (up positive)	free fall

Symbols: u = initial velocity, v = final velocity, a = acceleration (constant), t = time, s = displacement.

Part 1 — Vectors from Scratch (20 min)

Scalars and vectors (Hewitt §6.1, Glencoe §5.1)

A **scalar** has only a size (magnitude) and a unit. A **vector** has a size **and** a **direction**.

Scalars (size only)	Vectors (size + direction)
distance (m)	displacement (m, east)
speed (m/s)	velocity (m/s, north)
time (s), mass (kg)	acceleration (m/s ² , down)
temperature, energy	force (N, to the left)

How a vector is drawn: an **arrow**. Its **length** shows the size (to a chosen scale) and its **direction** shows the direction. In print a vector is written in bold (**A**) or with an arrow over the letter. Plain A means just its size.

Two useful facts:

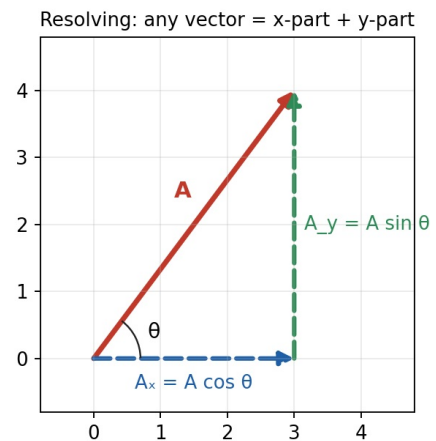
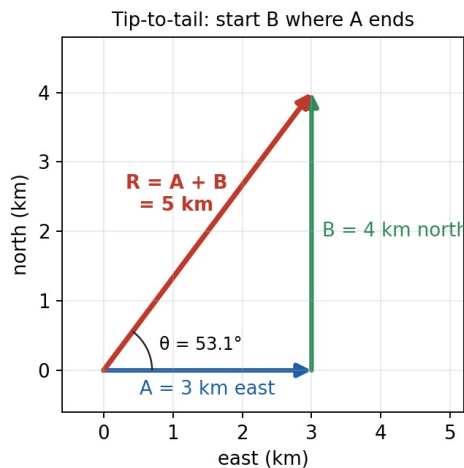
- Two vectors are **equal** if they have the same size and the same direction, wherever they are drawn.
- $-\mathbf{A}$ is the same size as $\mathbf{A}$ but points the opposite way. In one dimension, direction is just a **sign**: choose a positive direction (east, or up) and make the opposite way negative.

Adding vectors in one dimension

Along a line, add the signs. Walk 5 m east, then 2 m west, with east positive: $+5 + (-2) = +3$ m, so the displacement is 3 m east.

Adding vectors in two dimensions: tip-to-tail (Hewitt §6.4)

To add $\mathbf{A} + \mathbf{B}$, draw $\mathbf{A}$, then start $\mathbf{B}$ at the **tip** of $\mathbf{A}$. The **resultant** $\mathbf{R}$ runs from the tail of $\mathbf{A}$ to the tip of $\mathbf{B}$. The order doesn't matter: $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$.



Left: adding 3 km east and 4 km north tip-to-tail. Right: resolving a vector into components

When $\mathbf{A}$ and $\mathbf{B}$ are perpendicular they form a right triangle with $\mathbf{R}$ as the hypotenuse:

- **Size (Pythagoras):** $R = \sqrt{A^2 + B^2}$
- **Direction:** $\tan\theta = B/A$, so $\theta = \tan^{-1}(B/A)$, measured from $\mathbf{A}$.

Worked Example 1: the warm-up walk

3 km east then 4 km north.

- $R = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$ km.
- $\theta = \tan^{-1}(4/3) = 53.1^\circ$.

Answer: 5 km at 53.1° north of east. Distance walked = 7 km, but displacement = 5 km.

The range of possible answers. Two vectors of sizes A and B can give any resultant from $A - B$ (pointing opposite ways) to $A + B$ (the same way). For 3 and 4 the resultant can be anything from 1 to 7. It is 5 only when they are perpendicular.

Resolving a vector into components

Any vector can be replaced by two perpendicular vectors, its **components**, that add up to it. This is the reverse of adding. If $\mathbf{A}$ makes angle θ with the x-axis, then from the right triangle:

- $\cos\theta = \text{adjacent/hypotenuse} = A_x/A$, so $A_x = A \cos\theta$
- $\sin\theta = \text{opposite/hypotenuse} = A_y/A$, so $A_y = A \sin\theta$

Check: $A_x^2 + A_y^2 = A^2(\cos^2\theta + \sin^2\theta) = A^2$ ✓

Worked Example 2. A ball is kicked at 20 m/s at 30° above the horizontal. Find the components of its velocity.

- $v_x = 20 \cos 30^\circ = 20 \times 0.866 = 17.3 \text{ m/s}$ (horizontal)
- $v_y = 20 \sin 30^\circ = 20 \times 0.500 = 10.0 \text{ m/s}$ (vertical)

This is exactly how Session 2 splits a projectile's launch velocity.

Adding any vectors by components

For vectors that are **not** perpendicular, don't draw a messy triangle. Instead:

1. Resolve each vector into x and y components.
2. Add all the x's, and add all the y's: $R_x = A_x + B_x$, $R_y = A_y + B_y$.
3. Recombine: $R = \sqrt{R_x^2 + R_y^2}$ and $\theta = \tan^{-1}(R_y/R_x)$.

Worked Example 3. A boat moves 10 m east, then 6 m at 60° north of east. Find the total displacement.

- A: $A_x = 10$, $A_y = 0$.
- B: $B_x = 6 \cos 60^\circ = 3.0$, $B_y = 6 \sin 60^\circ = 5.20$.
- $R_x = 13.0 \text{ m}$, $R_y = 5.20 \text{ m}$.
- $R = \sqrt{(13.0^2 + 5.20^2)} = \sqrt{(169 + 27)} = \sqrt{196} = 14.0 \text{ m}$.
- $\theta = \tan^{-1}(5.20/13.0) = 21.8^\circ$.

Answer: 14.0 m at 21.8° north of east.

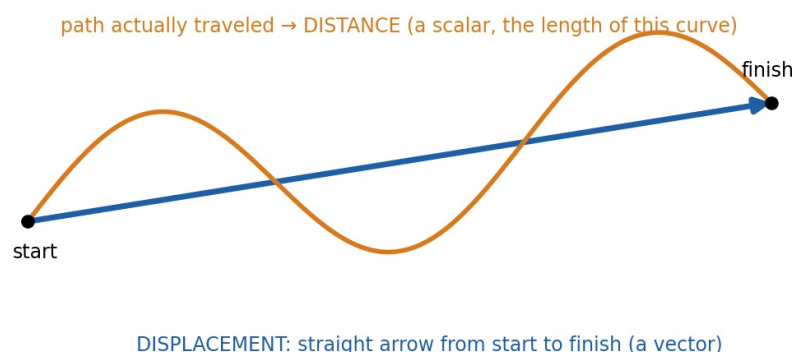
Part 2 — Describing Motion (10 min)

Motion is relative (Hewitt §2.1)

Every position and velocity is measured from a **reference point**. You sit still in a bus, but you are moving at 60 km/h relative to the road. Session 4 returns to this idea.

Distance vs displacement

- **Distance** is the length of the path actually traveled. It is a **scalar** and never negative.
- **Displacement** is the straight-line change in position, from start to finish. It is a **vector**: $\Delta x = x_{\text{final}} - x_{\text{initial}}$.



Distance follows the path; displacement is the straight arrow from start to finish

The size of the displacement is **never more** than the distance. The two are equal only for a straight-line trip in one direction. A runner who completes one lap of a 400 m track has run **400 m** but has a displacement of **zero**.

Speed vs velocity (Hewitt §2.2-2.3)

- **Average speed** = total distance / total time (scalar).

- **Average velocity** = displacement / time, $v_{\text{avg}} = \Delta x / \Delta t$ (vector).
- **Instantaneous** speed or velocity is the value at one moment: what the speedometer shows.

Units: m/s (SI), km/h, mph. To convert **km/h → m/s, divide by 3.6** (because 1 km/h = 1000 m / 3600 s).
For example, 90 km/h = 25 m/s.

Worked Example 4. A runner completes one lap of a 400 m track in 80 s. Find the average speed and average velocity.

- Average speed = $400/80 = 5.0$ m/s .
- Average velocity = $0/80 = 0$ m/s : back at the start, so the displacement is zero.

Acceleration (Hewitt §2.4, Glencoe §3.1)

Acceleration is the rate of change of **velocity**:

$$a = \Delta v / \Delta t = (v - u) / t, \text{ unit } \mathbf{m/s^2} \text{ ("meters per second, every second")}$$

Because velocity is a vector, you accelerate when you **speed up, slow down or change direction**.

Sign rule. If a and v point the **same** way, the object **speeds up**. If they point **opposite** ways, it **slows down**. "Negative acceleration" means "acceleration in the negative direction". It only means slowing down if the object is moving in the positive direction.

Part 3 — The Kinematic Equations (17 min)

These hold only when the **acceleration is constant**. Each is derived from the one before, so the student never has to memorize them blindly.

S1 from the definition of acceleration

Start from $a = (v - u) / t$. Multiply both sides by t : $at = v - u$. So:

$$v = u + at \quad (\text{S1})$$

The velocity changes by the same amount, a , every second, so v grows linearly with t .

S0: distance = average velocity × time

For any motion, displacement = average velocity × time. When a is constant, v rises in a straight line from u to v , so its average is exactly halfway:

$$v_{\text{avg}} = (u + v) / 2, \text{ so } s = \frac{1}{2}(u + v)t \quad (\text{S0})$$

This is the equation the student used first: "distance = average velocity × time".

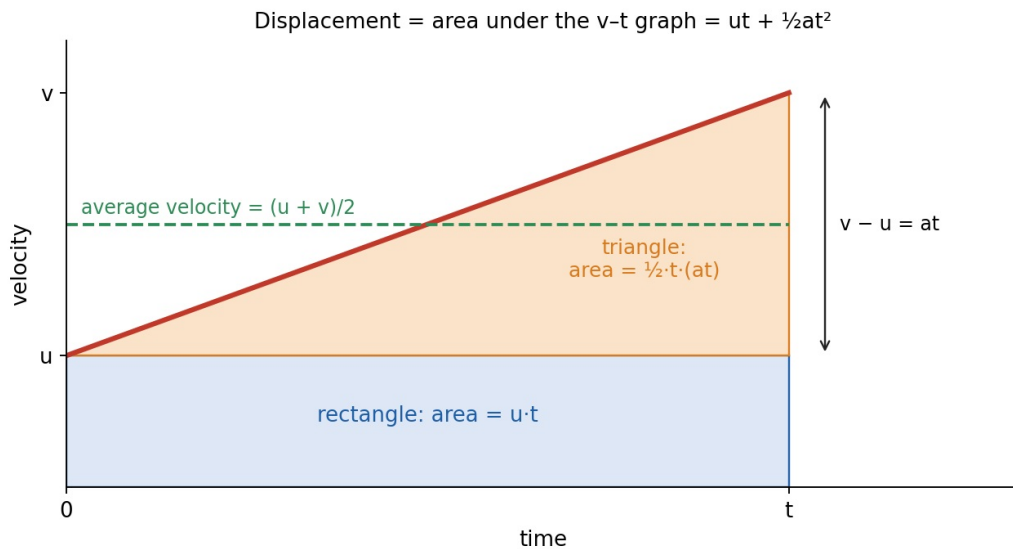
S2: substitute S1 into S0

Replace v in S0 with $u + at$:

$$s = \frac{1}{2}(u + u + at)t = \frac{1}{2}(2u + at)t = ut + \frac{1}{2}at^2$$

$$s = ut + \frac{1}{2}at^2 \quad (\text{S2})$$

The same result from the graph. The displacement is the **area under the v-t graph**. That area splits into a rectangle ($u \times t$) and a triangle ($\frac{1}{2} \times t \times at$):



The area under the velocity-time graph gives $s = ut + \frac{1}{2}at^2$

S3: eliminate time

Some problems give no time at all. From S1, $t = (v - u) / a$. Substitute into S0:

$$s = \frac{1}{2}(u + v) \times (v - u) / a = (v^2 - u^2) / (2a)$$

Multiply both sides by $2a$:

$$v^2 = u^2 + 2as \quad (\text{S3})$$

Choosing the equation

Each equation leaves out one of the five quantities. Pick the one that leaves out the quantity you are **neither given nor asked for**.

Equation	Leaves out	Use when...
S1 $v = u + at$	s	no distance involved
S0 $s = \frac{1}{2}(u + v)t$	a	no acceleration involved
S2 $s = ut + \frac{1}{2}at^2$	v	no final velocity involved
S3 $v^2 = u^2 + 2as$	t	no time involved

Method: (1) choose a positive direction; (2) list u , v , a , s , t with signs, writing “?” for the unknown; (3) pick the equation; (4) substitute and solve; (5) check the units and whether the answer is sensible.

Worked Example 5: speeding up

A car goes from rest to 25 m/s in 5.0 s. Find its acceleration and the distance covered.

- Known: $u = 0$, $v = 25 \text{ m/s}$, $t = 5.0 \text{ s}$.
- S1: $a = (25 - 0) / 5.0 = 5.0 \text{ m/s}^2$.
- S0: $s = \frac{1}{2}(0 + 25)(5.0) = 62.5 \text{ m}$. Check with S2: $0 + \frac{1}{2} \times 5.0 \times 25 = 62.5 \text{ m}$ ✓

Answer: 5.0 m/s^2 ; 62.5 m .

Worked Example 6: braking distance

A car moving at 30 m/s brakes with an acceleration of -6.0 m/s^2 . How far does it travel before stopping, and how long does it take?

- Forward is positive. Known: $u = 30$, $v = 0$, $a = -6.0$. Asked: s and t .
- S3: $0 = 30^2 + 2(-6.0)s$, so $s = 900/12 = 75$ m.
- S1: $0 = 30 + (-6.0)t$, so $t = 5.0$ s.

Answer: 75 m; 5.0 s. Double the speed and the stopping distance goes up **four** times ($s \propto u^2$). That is why speed limits matter.

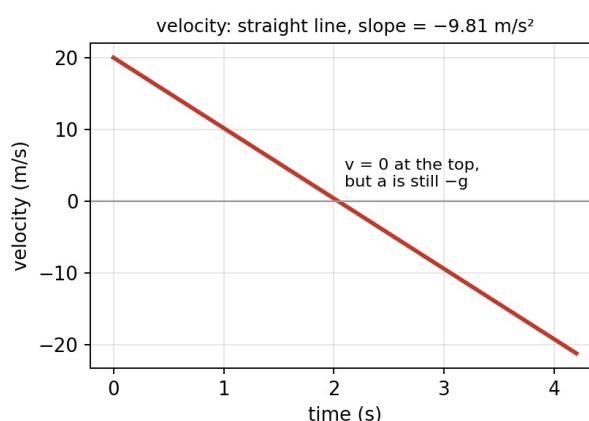
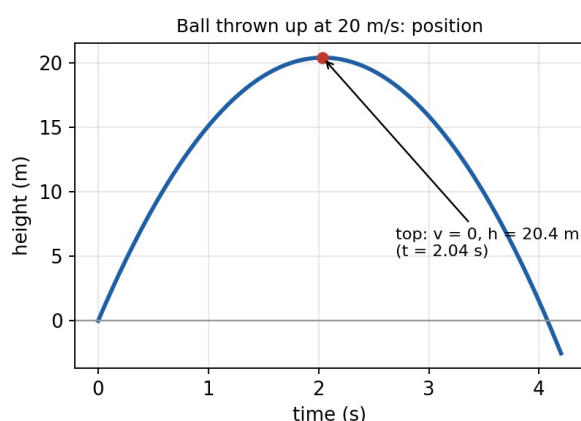
Part 4 — Free Fall (5 min)

Free fall (Hewitt §2.5-2.6, Glencoe §3.3)

Without air resistance, every object near Earth's surface accelerates downward at $g = 9.81 \text{ m/s}^2$, whatever its mass. Take **up as positive**, so $a = -9.81 \text{ m/s}^2$ in every free-fall problem, whether the object is going up, at the top, or coming down. The four equations are unchanged.

Useful results for an object thrown straight up at speed u :

- At the top, $v = 0$. From S1: $t_{\text{top}} = u/g$.
- From S3: $0 = u^2 - 2g \cdot h$, so $h_{\text{max}} = u^2/(2g)$.
- By symmetry it takes as long to fall as to rise, and it returns to your hand at the same speed it left with.



Ball thrown up at 20 m/s: position and velocity against time

Worked Example 7. A ball is thrown straight up at 15 m/s. Find (a) the maximum height, (b) the time to reach it, (c) the total time in the air.

- (a) $h = u^2/(2g) = 15^2/(2 \times 9.81) = 225/19.62 = 11.5$ m.
- (b) $t = u/g = 15/9.81 = 1.53$ s.
- (c) By symmetry, $2 \times 1.53 = 3.06$ s.

Exit Ticket (3 min)

1. Two forces of 6 N and 8 N act on an object. What are the largest and smallest possible resultants? → **14 N** (same direction) and **2 N** (opposite directions). 10 N only if they are perpendicular.
2. Which equation would you use to find how fast a car is going after 100 m, given u and a but not t ? → **S3**, $v^2 = u^2 + 2as$ (no time).
3. At the top of its flight, what are a ball's velocity and acceleration? → $v = 0$, but $a = 9.81 \text{ m/s}^2$ **downward**. It never stops accelerating.

Practice Problems

Use $g = 9.81 \text{ m/s}^2$. Conceptual questions C1–C7, numerical questions N1–N12. Do C3, N3 and N8 in class if time allows; the rest are homework.

Conceptual

- C1.** Sort into scalars and vectors: mass, displacement, speed, velocity, time, acceleration, force, distance, temperature.
- C2.** (a) Can the size of an object's displacement ever be greater than the distance it traveled? (b) Can the distance be non-zero while the displacement is zero? Give an example.
- C3.** Two vectors have sizes 6 units and 8 units. (a) What is the largest possible resultant? (b) The smallest? (c) At what angle between them is the resultant 10 units?
- C4.** Can an object have zero velocity and still be accelerating? Give an example.
- C5.** A car travels around a circular track at a steady 60 km/h. Is its (a) speed constant, (b) velocity constant, (c) acceleration zero? Explain.
- C6.** "A negative acceleration always means the object is slowing down." True or false? Give an example.
- C7.** Explain why the displacement in the equations of motion equals the area under the velocity–time graph.

Numerical

- N1.** A hiker walks 5 m east and then 12 m north. Find the size and direction of her displacement.
- N2.** A rope pulls a sled with a force of 50 N at 37° above the horizontal. Find the horizontal and vertical components of the force.
- N3.** A boat travels 10 m east and then 6 m at 60° north of east. Using components, find the resultant displacement. (Compare with Worked Example 3.)
- N4.** A family drives 120 km to a city in 1.5 h and returns home by the same road in 2.5 h. Find (a) the average speed for the round trip, (b) the average velocity for the round trip.
- N5.** Convert (a) 90 km/h to m/s, (b) 15 m/s to km/h.
- N6.** A cyclist speeds up uniformly from 4.0 m/s to 10 m/s in 3.0 s. Find (a) the acceleration, (b) the distance covered, using "distance = average velocity $\times$ time".
- N7.** A plane starts from rest and accelerates at 3.2 m/s^2 . It needs 80 m/s to take off. Find (a) the minimum runway length, (b) the time on the runway.
- N8.** A car moving at 20 m/s brakes at 4.0 m/s^2 . Find (a) the time to stop, (b) the stopping distance, (c) the distance covered in the first 2.0 s of braking.
- N9.** A stone is dropped from rest off a 45 m cliff. Find (a) the time to reach the bottom, (b) its speed on impact.
- N10.** A ball is thrown straight up at 20 m/s. Find (a) the maximum height, (b) the time to reach it, (c) its velocity after 3.0 s, (d) its height after 3.0 s.
- N11.** A car accelerates from rest at 2.0 m/s^2 for 10 s, then travels at constant velocity for 20 s, then brakes uniformly to a stop in 5.0 s. Find (a) the total distance, (b) the average velocity for the whole trip.
- N12.** A driver moving at 25 m/s sees a hazard. Her reaction time is 0.80 s, and then the car brakes at 5.0 m/s^2 . Find the total distance from seeing the hazard to stopping.

Answer Key — Full Step-by-Step Solutions

Conceptual

C1.

- **Scalars:** mass, speed, time, distance, temperature.
- **Vectors:** displacement, velocity, acceleration, force.

C2.

- **(a) No.** The straight line is the shortest path between two points, so the size of the displacement is at most the distance. They are equal only for straight-line motion in one direction.
- **(b) Yes.** One lap of a 400 m track: distance 400 m, displacement 0. The same goes for any round trip.

C3.

- **(a)** Same direction: $6 + 8 = 14$ units.
- **(b)** Opposite directions: $8 - 6 = 2$ units.
- **(c)** $6^2 + 8^2 = 36 + 64 = 100 = 10^2$, so by Pythagoras they must be at 90° .

C4. Yes. A ball thrown straight up has $v = 0$ for an instant at the top, but its acceleration is still 9.81 m/s^2 downward. That is why it immediately starts to fall.

C5.

- **(a) Yes,** the speed is constant (60 km/h).
- **(b) No.** Its direction keeps changing, so its velocity (a vector) changes.
- **(c) No.** A changing velocity means a non-zero acceleration. Session 4 shows it is v^2/r , toward the center.

C6. False. Negative acceleration means the acceleration points in the negative direction. A ball **falling** (taking up as positive) has negative velocity and negative acceleration, and it is **speeding up**. An object slows down only when a and v have opposite signs.

C7. Over a tiny time interval Δt the velocity is almost constant, so the displacement in that interval is $v \times \Delta t$: the area of a thin strip under the graph. Adding up all the strips gives the total area, and the total displacement. For constant acceleration the area is a rectangle ut plus a triangle $\frac{1}{2}at^2$, which is S_2 .

Numerical

N1.

- $R = \sqrt{(5^2 + 12^2)} = \sqrt{(25 + 144)} = \sqrt{169} = 13 \text{ m}.$
- $\theta = \tan^{-1}(12/5) = 67.4^\circ.$

Answer: 13 m at 67.4° north of east.

N2.

- Horizontal: $F_x = 50 \cos 37^\circ = 50 \times 0.799 = 39.9 \text{ N}.$
- Vertical: $F_y = 50 \sin 37^\circ = 50 \times 0.602 = 30.1 \text{ N}.$
- Check: $\sqrt{(39.9^2 + 30.1^2)} = \sqrt{(1592 + 906)} = \sqrt{2498} = 50.0 \text{ N} \checkmark$

Answer: 39.9 N horizontal; 30.1 N vertical.

N3.

- First vector: $(10, 0)$. Second: $(6 \cos 60^\circ, 6 \sin 60^\circ) = (3.0, 5.20)$.
- $R_x = 13.0 \text{ m}, R_y = 5.20 \text{ m}.$
- $R = \sqrt{(13.0^2 + 5.20^2)} = \sqrt{196} = 14.0 \text{ m}; \theta = \tan^{-1}(5.20/13.0) = 21.8^\circ.$

Answer: 14.0 m at 21.8° north of east. It matches Worked Example 3, as it should.

N4.

- **(a)** Total distance = $120 + 120 = 240 \text{ km}$; total time = $1.5 + 2.5 = 4.0 \text{ h}$. Average speed = $240/4.0 = 60 \text{ km/h}$.
- **(b)** Back home, so the displacement is 0. Average velocity = **0**.

Note that 60 km/h is **not** the average of 80 km/h and 48 km/h (64 km/h). The slower leg takes longer, so it counts for more.

N5.

- **(a)** $90/3.6 = 25 \text{ m/s}$.
- **(b)** $15 \times 3.6 = 54 \text{ km/h}$.

N6.

- **(a)** $a = (10 - 4.0)/3.0 = 2.0 \text{ m/s}^2$.
- **(b)** S0: $s = \frac{1}{2}(4.0 + 10)(3.0) = \frac{1}{2} \times 14 \times 3.0 = 21 \text{ m}$. Check with S2: $4.0 \times 3.0 + \frac{1}{2} \times 2.0 \times 3.0^2 = 12 + 9 = 21 \text{ m}$ ✓

Answer: 2.0 m/s²; 21 m.

N7.

- Known: $u = 0$, $v = 80 \text{ m/s}$, $a = 3.2 \text{ m/s}^2$.
- **(a)** S3: $80^2 = 0 + 2 \times 3.2 \times s$, so $s = 6400/6.4 = 1000 \text{ m}$ (1.0 km).
- **(b)** S1: $80 = 0 + 3.2t$, so $t = 25 \text{ s}$.

Answer: 1000 m; 25 s.

N8.

- Forward is positive: $u = 20$, $a = -4.0$.
- **(a)** S1: $0 = 20 - 4.0t$, so $t = 5.0 \text{ s}$.
- **(b)** S3: $0 = 20^2 - 2 \times 4.0 \times s$, so $s = 400/8.0 = 50 \text{ m}$.
- **(c)** S2 with $t = 2.0 \text{ s}$: $s = 20 \times 2.0 + \frac{1}{2} \times (-4.0) \times 2.0^2 = 40 - 8.0 = 32 \text{ m}$.

Answer: 5.0 s; 50 m; 32 m. More than half the distance is covered in the first 2 s, because the car is fastest then.

N9.

- Take down as positive to keep the signs simple: $u = 0$, $a = +9.81$, $s = 45 \text{ m}$.
- **(a)** S2: $45 = 0 + \frac{1}{2} \times 9.81 \times t^2$, so $t^2 = 90/9.81 = 9.17$ and $t = 3.03 \text{ s}$.
- **(b)** S3: $v^2 = 0 + 2 \times 9.81 \times 45 = 883$, so $v = 29.7 \text{ m/s}$. Check with S1: $9.81 \times 3.03 = 29.7$ ✓

Answer: 3.03 s; 29.7 m/s.

N10.

- Up is positive: $u = 20$, $a = -9.81$.
- **(a)** S3 with $v = 0$ at the top: $0 = 20^2 - 2 \times 9.81 \times h$, so $h = 400/19.62 = 20.4 \text{ m}$.
- **(b)** S1: $0 = 20 - 9.81t$, so $t = 2.04 \text{ s}$.
- **(c)** S1 at $t = 3.0 \text{ s}$: $v = 20 - 9.81 \times 3.0 = 20 - 29.4 = -9.43 \text{ m/s}$. The minus sign means it is moving **downward** at 9.43 m/s.
- **(d)** S2: $s = 20 \times 3.0 - \frac{1}{2} \times 9.81 \times 3.0^2 = 60 - 44.1 = 15.9 \text{ m}$ above the launch point.

Answer: 20.4 m; 2.04 s; 9.43 m/s downward; 15.9 m.

N11.

- **Stage 1:** $u = 0$, $a = 2.0$, $t = 10$. $v = 20 \text{ m/s}$; $s_1 = \frac{1}{2} \times 2.0 \times 10^2 = 100 \text{ m}$.

- **Stage 2:** constant 20 m/s for 20 s: $s_2 = 20 \times 20 = 400 \text{ m}$.
- **Stage 3:** 20 $\rightarrow$ 0 m/s in 5.0 s. S0: $s_3 = \frac{1}{2}(20 + 0)(5.0) = 50 \text{ m}$.
- **(a) Total:** $100 + 400 + 50 = 550 \text{ m}$.
- **(b)** It moves in one direction, so the displacement is 550 m. Total time: $10 + 20 + 5.0 = 35 \text{ s}$.
 $v_{\text{avg}} = 550/35 = 15.7 \text{ m/s}$.

Answer: 550 m; 15.7 m/s.

N12.

- **Reaction phase** (constant speed, no braking yet): $s_1 = 25 \times 0.80 = 20 \text{ m}$.
- **Braking phase:** $0 = 25^2 - 2 \times 5.0 \times s_2$, so $s_2 = 625/10 = 62.5 \text{ m}$.
- **Total:** $20 + 62.5 = 82.5 \text{ m}$.

Answer: 82.5 m. The reaction distance alone is about a quarter of the total, which is why distracted driving is so dangerous.

Notes for next session

Session 2 takes these tools into **two dimensions**. A projectile's launch velocity is resolved into components (V2). The horizontal motion is constant velocity, and the vertical motion is free fall using S1-S3 with $a = -g$. The key idea: **the two directions are independent**. The warm-up for Session 2 asks for these equations from memory.